

Answer Key: Warm-Up on $Ax = b$

1. Matrix form.

System A: $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$

System B: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1.1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 2.1 \end{bmatrix}.$

2. Solutions.

Both systems have solution $(x, y) = (1, 1).$

- A: adding the equations gives $2x = 2$, so $x = 1$, then $y = 1$.
- B: subtracting the first equation from the second gives $0.1y = 0.1$, so $y = 1$, then $x = 1$.

3. After the 0.1 change in $\mathbf{b}$.

- A with $\mathbf{b} = (2, 0.1)$: $2x = 2.1$, so $x = 1.05$ and $y = 0.95$.
- B with $\mathbf{b} = (2, 2.2)$: $0.1y = 0.2$, so $y = 2$ and $x = 0$.

4. Sketches. In A the lines $y = 2 - x$ and $y = x$ cross at a healthy right angle. In B the lines $y = 2 - x$ and $y = (2.1 - x)/1.1$ have slopes -1 and about -0.91 , so they are almost parallel and nearly on top of each other. Sliding one line by a small amount slides the crossing point a long way along the other.

5. Size of the change. System A moved by 0.05 in each coordinate. System B moved by 1 in x and 1 in y , which is 20 times as much, from the same 0.1 nudge to $\mathbf{b}$. Accept any explanation that points to the lines being nearly parallel in B.

6. Trust. System A. A small error in $\mathbf{b}$ barely changes its answer, while in B the same error gets amplified into a large error in $\mathbf{x}$.

Vocabulary to introduce afterward

A matrix like the one in System B is called **ill-conditioned**: it is close to singular (the lines are nearly parallel), so small changes in $\mathbf{b}$ produce large changes in $\mathbf{x}$. System A's matrix is **well-conditioned**.

Optional extension (teacher only)

`np.linalg.cond(A)` returns 1 for System A and about 42 for System B. The determinants have magnitude 2 and 0.1, which hints at the same story, but determinant alone is a rough guide because rescaling a matrix changes it without changing the conditioning.

Tie to the course. When two columns of a feature matrix X are nearly parallel (correlated features), $X^T X$ is ill-conditioned, and the normal equations behave like System B. That is why regression coefficients become unstable with highly correlated features.